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Clinical landscape of therapeutic vaccines for ovarian cancer: current status, challenges, and future directions

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Abstract

Background Therapeutic cancer vaccines have emerged as a promising immunotherapeutic strategy for ovarian cancer by stimulating tumor-specific immune responses and establishing long-term immune memory. Despite decades of clinical investigation, the overall development landscape, translational characteristics, and research trends of ovarian cancer vaccines remain incompletely understood.

Methods Clinical trials related to ovarian cancer vaccines were retrieved from the ClinicalTrials.gov database. A total of 136 unique studies registered between 1996 and June 2026 were included. Information regarding study phase, recruitment status, enrollment size, geographic distribution, funding source, vaccine platform, combination strategy, clinical application context, and endpoint selection was extracted and descriptively analyzed. Relationships among vaccine platforms, combination strategies, clinical phases, and endpoint categories were further visualized. Immune monitoring outcomes were further coded according to reporting specificity, assay-method class, and biological domain.

Results A total of 136 clinical studies involving 4,180 participants were identified. Most studies were early-phase trials, including Phase I (48.53%), Phase II (23.53%), and Phase I/II (21.32%), whereas only 2 Phase III studies were found. Completed studies accounted for 55.88%, while only 22.79% had publicly posted results. The United States dominated the research landscape, participating in 105 studies. Major vaccine platforms included peptide/antigen vaccines, dendritic-cell vaccines, whole tumor cell or lysate vaccines, DNA/RNA vaccines, and viral/vector vaccines. Current development trends indicate a transition from single-vaccine approaches toward combination strategies involving immune checkpoint inhibitors, chemotherapy, and immune adjuvants. Most studies focused on advanced, metastatic, or recurrent ovarian cancer. Safety/toxicity and immunogenicity were the most frequently used endpoints, whereas progression-free survival, overall survival, quality of life, and biomarker-based outcomes were comparatively underrepresented. Immune monitoring or immune-response outcomes were reported in 92 trials (67.65%), but only 58 trials (42.65%) specified an assay method; 34 trials (25.00%) described an immune domain without a named assay.

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Conclusions Therapeutic vaccine development for ovarian cancer has shown sustained progress over the past three decades but remains largely in an early exploratory stage. Although diverse vaccine platforms and combination immunotherapy strategies demonstrate substantial translational potential, confirmatory evidence remains limited. Future efforts should prioritize biomarker-guided patient selection, personalized vaccine development, standardized immune monitoring, clinically meaningful endpoints, and large-scale multicenter randomized Phase II/III trials to facilitate the translation of vaccine-induced immune responses into tangible clinical benefits. The observed heterogeneity in immune-response reporting further highlights the need for prespecified, harmonized, and vaccine-specific immune monitoring frameworks in future vaccine trials.

Keywords Ovarian cancer, Therapeutic vaccines, Immunotherapy, Clinical trial landscape, ClinicalTrials.gov, Combination immunotherapy, Personalized vaccines, Translational research

Introduction

Ovarian cancer is a highly heterogeneous malignancy arising from the ovary, fallopian tube, or primary peritoneum. Epithelial ovarian cancer accounts for the majority of cases, and high-grade serous ovarian cancer is the most common and most aggressive subtype [1]. Molecular pathological studies have shown that ovarian cancer is not a single disease entity, but rather a complex disease spectrum shaped by distinct histological subtypes, molecular driver events, and the tumor microenvironment. High-grade serous ovarian cancer is frequently characterized by TP53 mutation, homologous recombination repair deficiency, and alterations in DNA damage repair genes such as BRCA1/2. These molecular features not only contribute to tumor initiation and progression, but also determine patient eligibility for precision strategies such as PARP inhibitor therapy [1, 2]. Epidemiologically, although ovarian cancer is less common than breast, cervical, and endometrial cancers, it remains one of the leading causes of death among gynecologic malignancies. GLOBOCAN 2022 data indicate that ovarian cancer continues to represent an important cause of cancer-related mortality among women worldwide [3]. Because early symptoms are nonspecific and effective screening strategies are lacking, most patients are diagnosed at an advanced stage. Peritoneal dissemination, ascites formation, recurrence, and treatment resistance are key determinants of poor prognosis.

The progression of ovarian cancer is driven not only by intrinsic genomic abnormalities in tumor cells, but also by dynamic interactions with the tumor immune microenvironment. Ovarian cancer tissues contain tumor-infiltrating lymphocytes, antigen-presenting cells, regulatory T cells, myeloid-derived suppressor cells, tumor-associated macrophages, and multiple immune checkpoint molecules. Previous studies have demonstrated that intratumoral T-cell infiltration is associated with improved clinical outcomes in epithelial ovarian cancer, suggesting that host immunity plays an important role in disease control [4]. However, ovarian cancer cells and their microenvironment can evade immune surveillance through activation of the PD-1/PD-L1 axis, suppression

of effector T-cell function, and enrichment of immunosuppressive cell populations [5]. This state of immune recognition with insufficient immune effector activity provides a theoretical basis for immunotherapy and therapeutic cancer vaccines, and may also explain why single-agent immune interventions have shown limited efficacy in ovarian cancer.

At present, the standard treatment for ovarian cancer remains cytoreductive surgery combined with platinum- and taxane-based chemotherapy. For newly diagnosed advanced ovarian cancer, the extent of cytoreduction, platinum sensitivity, and subsequent maintenance therapy are central determinants of prognosis. In recent years, PARP inhibitors and anti-angiogenic therapy have substantially changed the landscape of first-line maintenance treatment. Long-term follow-up of SOLO1 showed that olaparib as first-line maintenance therapy provides durable benefit in patients with BRCA-mutated advanced ovarian cancer [6]. The PRIMA and PAOLA-1 trials further supported the use of PARP inhibitors in homologous recombination-deficient and broader patient populations [7, 8]. In addition, the antibody-drug conjugate mirvetuximab soravtansine has provided a new therapeutic option for patients with folate receptor alpha-positive, platinum-resistant ovarian cancer [9]. Nevertheless, most patients with advanced disease still experience relapse after initial treatment and eventually develop platinum-resistant disease. PARP inhibitors, anti-angiogenic agents, and chemotherapy are all limited by resistance, toxicity, and restricted applicability. Moreover, single-agent immune checkpoint inhibitors have shown limited objective response rates in unselected ovarian cancer populations [10]. Therefore, strategies capable of inducing more durable, specific, and memory-based antitumor immunity remain an important direction in ovarian cancer research.

Therapeutic cancer vaccines are active immunotherapeutic strategies designed to activate antitumor immune responses by presenting tumor-associated antigens or tumor-specific antigens. Their core mechanisms include uptake and processing of tumor antigens by antigen-presenting cells, presentation via major histocompatibility

complex molecules, induction of CD8 + cytotoxic T cells, CD4 + helper T cells, humoral immune responses, and the formation of immunological memory [11, 12]. Unlike passive immunotherapies, cancer vaccines aim to mobilize the patient's own immune system to recognize and eliminate tumor cells. Theoretically, they may be particularly suitable for minimal residual disease, maintenance therapy, recurrence prevention, and low tumor burden settings. This is because vaccine-induced antitumor immunity requires sequential processes, including antigen uptake and presentation by antigen-presenting cells, activation and expansion of tumor-specific T cells, and establishment of long-term immune memory [13]. These immune responses may be more effective when tumor burden is limited, whereas advanced tumors with a high tumor burden are often associated with profound immunosuppressive mechanisms, including T-cell dysfunction, increased regulatory immune cell populations, and impaired antigen presentation, which can restrict vaccine-induced antitumor activity [14]. With advances in neoantigen prediction, mRNA platforms, dendritic-cell engineering, and combination immunotherapy, cancer vaccines have re-emerged as an important field in tumor immunotherapy [11, 12].

In ovarian cancer, vaccine platforms that have been explored include peptide or antigen vaccines, dendritic-cell vaccines, whole tumor cell or tumor lysate vaccines, DNA/RNA vaccines, viral vector vaccines, and personalized neoantigen vaccines. Common targets include p53, NY-ESO-1, WT1, MUC1, HER2, mesothelin, folate receptor alpha, CA-125, and survivin. Each platform has distinct features. Peptide or antigen vaccines are relatively easy to standardize and monitor immunologically, but are constrained by antigen expression and HLA restriction. Dendritic-cell vaccines have strong antigen-presenting capacity but require complex manufacturing. Whole tumor lysate vaccines may cover a broader antigen spectrum, but standardization and mechanistic interpretation are more challenging. DNA/RNA vaccines and personalized neoantigen vaccines offer potential advantages in rapid design and individualized adaptation. Previous ovarian cancer vaccine studies have shown that dendritic-cell vaccines and whole tumor lysate-based strategies can induce tumor antigen-specific immune responses, although clinical benefit requires confirmation in higher-quality studies [15]. More recently, ovarian cancer vaccine research has shifted from monotherapy toward combination strategies, including combinations with adjuvants such as GM-CSF, chemotherapy, immune checkpoint inhibitors, PARP inhibitors, or anti-angiogenic therapy, with the goal of overcoming the immunosuppressive tumor microenvironment and enhancing vaccine-induced T-cell responses [11, 12, 15].

Although cancer vaccines for ovarian cancer have a clear immunological rationale and translational potential, important evidence gaps remain. Previous clinical studies have largely been concentrated in phase I or phase I/II settings, with small sample sizes and substantial heterogeneity in study design, vaccine platform, antigen selection, combination regimen, and endpoint definition. Many studies have used safety and immunogenicity as primary endpoints, and whether these immune responses can translate into progression-free survival, overall survival, or objective response benefits remains insufficiently established. Patient selection and biomarker stratification are also inadequate, and it remains unclear which patients are most likely to benefit. In addition, existing reviews have often focused on specific vaccine platforms or immunotherapeutic mechanisms, whereas systematic descriptions of the overall clinical development landscape of ovarian cancer vaccines based on clinical trial registry data remain limited. Therefore, this study used the ClinicalTrials.gov database to systematically characterize the clinical research landscape of cancer vaccines for ovarian cancer, including temporal trends, trial phases, study status, geographic distribution, technological routes, combination strategies, endpoint selection, and active pipelines, with the aim of informing future clinical trial design and translational research.

Materials and methods

To systematically evaluate the clinical development status and research trends of cancer vaccines for ovarian cancer, this study conducted a descriptive analysis based on publicly available registration data from ClinicalTrials.gov. ClinicalTrials.gov is a clinical trial registry platform maintained by the U.S. National Library of Medicine and includes clinical studies registered by investigators, academic institutions, industry sponsors, and governmental organizations worldwide [16]. This study focused on ovarian cancer-related cancer vaccine clinical studies. Trial registration records were retrieved and exported, and information on study phase, status, vaccine platform, combination strategy, endpoint setting, funding type, enrollment size, and geographic distribution was systematically extracted and analyzed.

The data source was an ovarian cancer-related cancer vaccine clinical trial dataset exported from ClinicalTrials.gov. At the time of data export, 136 unique clinical study records were identified. The extracted fields included NCT number, study title, study URL, recruitment status, brief summary, availability of posted results, disease condition, intervention, primary outcome measures, secondary outcome measures, sponsor, collaborators, sex, age, study phase, enrollment size, funding type, study type, study design, start date, primary completion date, completion date, first posted date, last update posted date,

study locations, and study documents. All records were deduplicated and verified using the NCT number as the unique identifier.

A critical note on trial counting is necessary. In this study, the total number of studies refers to unique clinical trial records after deduplication by NCT number ($N=136$). However, when analyzing geographic distribution, a single multicenter or multinational trial may be conducted in multiple countries; therefore, each participating country was counted as one country participation record. As a result, the sum of country-level counts may exceed the number of unique trials, reflecting the real-world nature of multicenter and multinational clinical research. Similarly, vaccine platform, combination strategy, and endpoint category analyses used non-mutually exclusive classifications because a single study could involve multiple vaccine platforms, combination components, or endpoint categories. Therefore, the sums of these categories may also exceed 136.

Key variables were standardized before analysis. Study phase was categorized according to the original ClinicalTrials.gov field as Early Phase I, Phase I, Phase I/Phase II, Phase II, Phase II/Phase III, Phase III, or NA. Study status was categorized as Completed, Terminated, Withdrawn, Recruiting, Active not recruiting, Not yet recruiting, or Unknown. Funding type was classified according to the Funder Type field as Other, Industry, NIH, Network, or FED. Vaccine platforms were manually assisted and classified based on keywords in study titles, summaries, conditions, and interventions, including peptide/antigen vaccines, whole tumor cell or tumor lysate vaccines, dendritic-cell vaccines, DNA/RNA vaccines, viral/vector vaccines, and vaccine-related studies that could not be clearly classified. Combination strategies were further categorized as GM-CSF/adjuvant combination, chemotherapy or standard-of-care combination, immune checkpoint inhibitor combination, or no clearly defined combination strategy. Endpoint types were classified according to primary, secondary, and other outcome fields as safety/toxicity, immune response, objective response or radiographic response, overall survival, progression-free survival, CA-125, and quality of life.

To address heterogeneity in immune-response assessment, immune monitoring outcomes were additionally coded at the study level. Reporting status was categorized as explicit assay/method reported, immune domain reported but assay not specified, or no immune monitoring outcome reported. Explicit assay classes included ELISA-based assays, ELISpot, flow cytometry/ICS/FACS, tetramer/MHC multimer, cytokine/soluble-factor measurement, T-cell proliferation/expansion, CTL/cytotoxicity/51Cr, immune-cell phenotyping/TIL/IHC, molecular immune profiling, and DTH/skin testing. Biological domains were coded as humoral/

antibody response, T-cell/cellular response, cytokine/soluble-factor response, immune-cell phenotype/infiltration, local/DTH response, immune memory/durability, or generic immune response. Assay-method and biological-domain categories were non-mutually exclusive, and clinical immune-response criteria wording such as irRC/iRECIST was flagged separately rather than counted as immune assay monitoring.

Data processing and statistical analyses were primarily performed using Microsoft Excel. Categorical variables were described using frequencies and percentages. Continuous variables such as enrollment size were summarized using total, median, and maximum values. Temporal trends were analyzed according to study start year and first posted year. Country information was extracted from the Locations field to visualize geographic distribution. Sankey diagrams were constructed to illustrate relationships among vaccine platform, combination strategy, and study phase, and chord diagrams were generated to show relationships between endpoint type and clinical phase. All figures were based on publicly available registry data and were designed to visualize the clinical development landscape of cancer vaccines for ovarian cancer.

All data used in this study were obtained from publicly accessible ClinicalTrials.gov registration records. The study did not involve identifiable individual-level patient data, interventional procedures, or human biological samples. Therefore, ethics committee approval and informed consent were not required.

Results

Overall characteristics of clinical trials

As of June 1, 2026, 136 clinical studies related to cancer vaccines for ovarian cancer were identified in ClinicalTrials.gov. The study start years ranged from 1996 to 2026, indicating that this field has been under clinical investigation for nearly three decades. Across all studies, the total planned or actual enrollment was 4,180 patients, with a median enrollment of 20.5 and a maximum enrollment of 214, suggesting that the field is largely characterized by small-sample exploratory studies.

Regarding annual trends, the number of ovarian cancer vaccine clinical studies fluctuated over time rather than increasing linearly. Trial registration was relatively concentrated between 2010 and 2012, reaching a peak of 10 studies in 2011. A smaller peak was observed in 2017, with 9 studies. New studies have continued to be initiated in recent years, indicating ongoing clinical exploration of cancer vaccines in ovarian cancer (Fig. 1A).

Distribution of clinical phases

The phase distribution showed that ovarian cancer vaccine studies were predominantly concentrated in early clinical development. Phase I studies accounted for 66

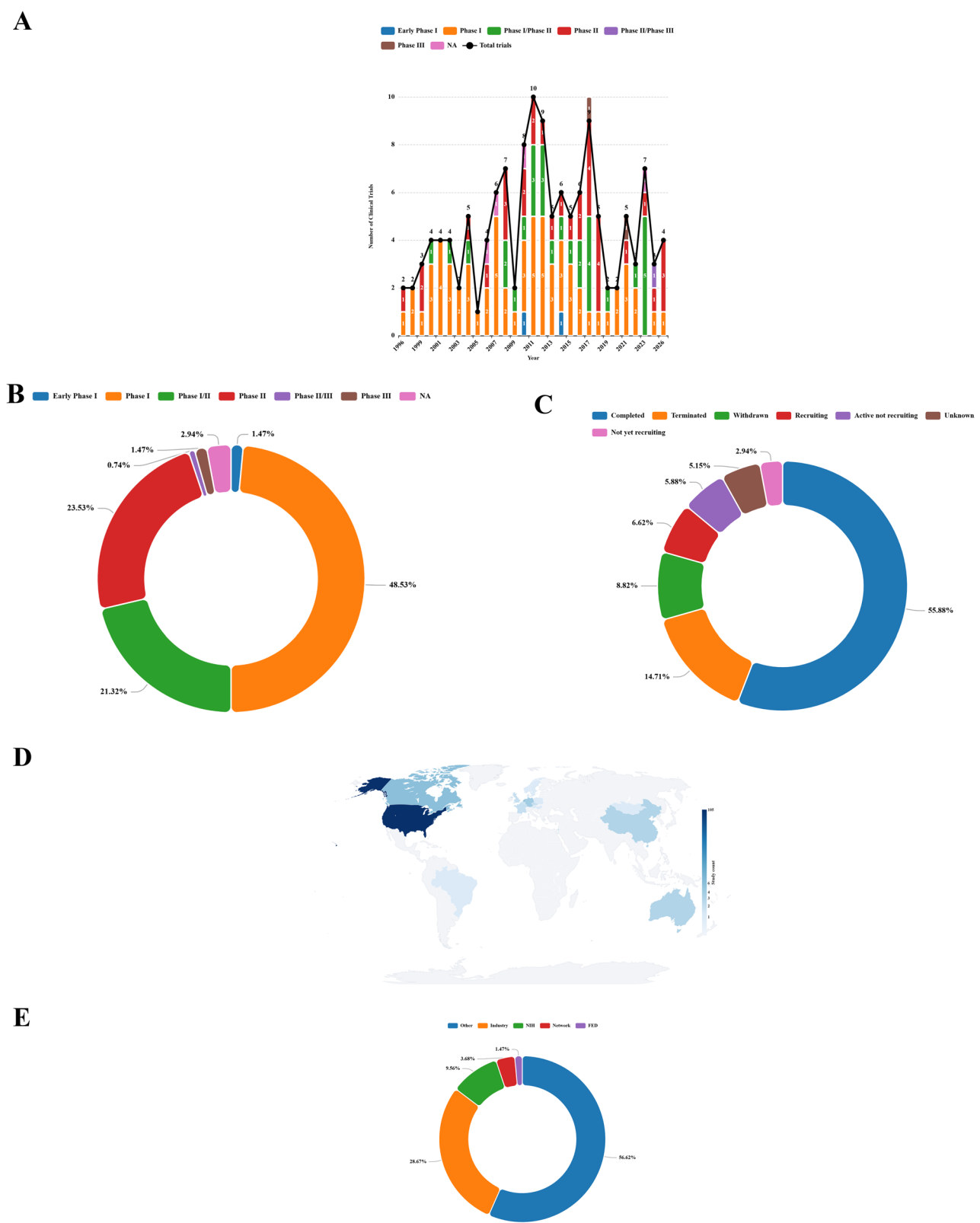


Fig. 1 Overall landscape of ovarian cancer vaccine clinical trials. **(A)** Annual number of trials by start year and clinical phase from 1996 to 2026. **(B)** Distribution of clinical phases. **(C)** Distribution of recruitment status. **(D)** Geographic distribution of participating countries. **(E)** Distribution of funding types. A total of 136 unique ClinicalTrials.gov records were included. NA, not applicable; NIH, National Institutes of Health; FED, U.S. federal funding

trials (48.53%), Phase II studies for 32 trials (23.53%), and Phase I/Phase II studies for 29 trials (21.32%). In contrast, only 2 Phase III studies (1.47%) and 1 Phase II/Phase III study (0.74%) were identified. Early Phase I studies accounted for 2 trials (1.47%), and 4 studies were classified as NA (2.94%) (Fig. 1B). These findings suggest that the clinical development of ovarian cancer vaccines remains focused mainly on safety assessment, dose exploration, immunogenicity validation, and preliminary efficacy evaluation, with relatively few studies entering confirmatory phases.

Study status, result disclosure, and the current active pipeline

In terms of study status, 76 studies (55.88%) were completed, representing the largest category. Twenty studies (14.71%) were terminated, and 12 studies (8.82%) were withdrawn. Studies that remained clinically active included 9 Recruiting studies, 8 Active not recruiting studies, and 4 Not yet recruiting studies, totaling 21 studies (15.44%). Seven studies (5.15%) had an Unknown status (Fig. 1C). In addition, registry information showed that only 31 studies had posted results, whereas 105 studies had not, suggesting a degree of underreporting or insufficient public disclosure of results in this field.

Table 1 further summarizes representative active or ongoing ovarian cancer vaccine studies. These studies include peptide/antigen vaccines, dendritic-cell vaccines, whole tumor cell or tumor lysate vaccines, DNA/RNA vaccines, viral/vector vaccines, and vaccine strategies related to engineered cell therapy. Some projects have relatively large planned enrollment sizes. For example, the RNA vaccine plus cell therapy study NCT04503278 plans to enroll 214 patients, and the KRAS-reactive engineered T-cell plus vaccine study NCT06253520 plans to enroll 210 patients, suggesting that personalized or molecularly targeted vaccine strategies are emerging as important directions. Meanwhile, several ongoing studies use combination approaches, such as NeoVax plus nivolumab, DPX-Survivac plus pembrolizumab and cyclophosphamide, and p53MVA plus pembrolizumab, reflecting a transition from single-vaccine strategies toward combined immune modulation.

Geographic distribution and funding sources

Geographic distribution showed that clinical studies of cancer vaccines for ovarian cancer were highly concentrated in the United States. By participating country, the United States was involved in 105 studies, far exceeding all other countries. Canada and the Netherlands were each involved in 6 studies, Germany in 4 studies, China and Australia in 3 studies each, and France, the United Kingdom, Czechia, Belgium, and Israel in 2 studies each. Other countries had only a small number of records

(Fig. 1D). This distribution indicates that ovarian cancer vaccine development has been driven primarily by research centers in North America and parts of Europe, with an uneven global distribution of research resources.

Regarding funding type, the Other category was the most common, comprising 77 studies (56.62%). Industry funded 39 studies (28.67%), NIH funded 13 studies (9.56%), and Network and FED funding accounted for 5 and 2 studies, respectively (Fig. 1E). Table 2 further shows that the National Cancer Institute, University of Washington, Roswell Park Cancer Institute, Memorial Sloan Kettering Cancer Center, Abramson Cancer Center at Penn Medicine, Gradalis, and Ludwig Institute were among the more active sponsors in this field. The National Cancer Institute was involved in 13 studies and the University of Washington in 9 studies, indicating that academic institutions, governmental research organizations, and research-oriented cancer centers play central roles in the clinical development of ovarian cancer vaccines.

Relationships among vaccine platforms, combination strategies, and clinical phases

Vaccine platforms and combination strategies were markedly heterogeneous. According to Table 3, major ovarian cancer vaccine platforms included peptide/antigen vaccines, whole tumor cell or tumor lysate vaccines, dendritic-cell vaccines, DNA/RNA vaccines, and viral/vector vaccines. Peptide/antigen vaccines are relatively amenable to standardization and immune monitoring, with common targets including p53, NY-ESO, folate receptor alpha, survivin, mesothelin, and IGFBP-2. Whole tumor cell or lysate vaccines provide broader antigen coverage but involve more complex manufacturing and quality control. Dendritic-cell vaccines emphasize antigen presentation and are often combined with GM-CSF, checkpoint inhibitors, or local therapeutic approaches. DNA/RNA vaccines and viral vector vaccines represent emerging directions with greater potential for personalization and platform-based development.

Figure 2A illustrates the relationships among vaccine category, combination strategy, and clinical phase. It should be noted that the flows in the Sankey diagram represent non-mutually exclusive classification paths rather than unique trial counts. Because a single study may involve multiple vaccine platforms or combination strategies, the sum of flows on the right side may exceed 136. Overall, peptide/antigen vaccines and whole tumor cell/lysate vaccines were the major categories connected to Phase I, Phase II, and Phase I/Phase II studies. Combination strategies included no clearly defined combination strategy, GM-CSF/adjuvant combinations, chemotherapy/standard-of-care combinations, and immune checkpoint inhibitor combinations, indicating that the field is

Table 1 Representative active or ongoing ovarian cancer vaccine trials

NCT Number	Study title	Current status	Phase	Enrollment	Vaccine platform	Combination strategy	Clinical setting	End-point focus	Sponsor	Primary completion date
NCT04024878	NeoVax With Nivolumab in Patients With Ovarian Cancer	Active Not Recruiting	Phase 1	22	Not clearly specified	Checkpoint inhibitor	Not clearly specified	Safety/tolerability; Immune response; Radio-graphic response; Survival	Dana-Farber Cancer Institute	2025-04-30
NCT04503278	A Clinical Study of the Safety and Effectiveness of an Investigational Cell Therapy Given With and Without ...	Active Not Recruiting	Phase 1	214	DNA/RNA	Not clearly specified	Recurrent disease; Advanced/metastatic; Basket/multiple solid tumors	Safety/tolerability; Immune response; Radio-graphic response	BioN-Tech Cell & Gene Therapies GmbH	2028-08-01
NCT06253520	Autologous T-cells Genetically Engineered to Express Receptors Reactive Against KRAS Mutations in Conjunction...	Active Not Recruiting	Phase 1	210	Not clearly specified	Chemotherapy/SOC	Advanced/metastatic; Maintenance/adjuvant prevention; Basket/multiple solid tumors	Safety/tolerability; Immune response; Radio-graphic response	National Cancer Institute (NCI)	2029-02-20
NCT00799110	Vaccination of Patients With Ovarian Cancer With Dendritic Cell/Tumor Fusions With Granulocyte Macrophage C...	Active Not Recruiting	Phase 2	23	Dendritic cell/DC	Adjuvant/GM-CSF	Not clearly specified	Safety/tolerability; Immune response; Radio-graphic response	Beth Israel Deaconess Medical Center	2024-12-01
NCT02346747	A Trial of Vigil for Participants With Ovarian Cancer	Active Not Recruiting	Phase 2	92	Not clearly specified	Not clearly specified	Recurrent disease; Basket/multiple solid tumors	Radio-graphic response; Survival	Gradalis, Inc.	2028-12-01
NCT03029403	Phase 2 Study of Pembrolizumab, DPX-Survivac Vaccine and Cyclophosphamide in Advanced Ovarian, Primary Peri...	Active Not Recruiting	Phase 2	47	Not clearly specified	Checkpoint inhibitor; Chemotherapy/SOC	Advanced/metastatic; Maintenance/adjuvant prevention; Basket/multiple solid tumors	Safety/tolerability; Immune response; Radio-graphic response; Survival	University Health Network, Toronto	2025-12-12

Table 1 (continued)

NCT Number	Study title	Current status	Phase	Enrollment	Vaccine platform	Combination strategy	Clinical setting	End-point focus	Sponsor	Primary completion date
NCT03113487	P53MVA and Pembrolizumab in Treating Patients With Recurrent Ovarian, Primary Peritoneal, or Fallopian Tube...	Active Not Recruiting	Phase 2	29	Peptide/antigen; Whole tumor/cell lysate; Viral/vector	Checkpoint inhibitor	Recurrent disease; Platinum-resistant	Immune response; Radio-graphic response; Survival	City of Hope Medical Center	2022-08-26
NCT04713514	OSE2101 Alone or in Combination With Pembrolizumab vs. BSC in Patient With Platinum-sensitive Recurrent OC	Active Not Recruiting	Phase 2	180	DNA/RNA	Checkpoint inhibitor; Chemotherapy/SOC	Recurrent disease; Platinum-sensitive; Maintenance/adjuvant prevention	Safety/tolerance; Radio-graphic response; Survival	AR-CAGY/GINECO GROUP	2025-12-01
NCT05479045	A Combination Therapy Strategy to Prevent Anti-PD-1 Therapy Resistance in Metastatic Ovarian Cancer Patients	Not Yet Recruiting	Phase 2	24	Peptide/antigen	Checkpoint inhibitor	Advanced/metastatic; Maintenance/adjuvant prevention	Safety/tolerance; Radio-graphic response; Survival	Georgetown University	2027-11-01
NCT05794659	Adjuvant Therapeutic Cancer Vaccine (AST-201, pUMVC3-hIGFBP-2) in Patients With Advanced Ovarian Cancer	Not Yet Recruiting	Phase 2	98	Peptide/antigen	Adjuvant/GM-CSF; Chemotherapy/SOC	Advanced/metastatic; Maintenance/adjuvant prevention	Safety/tolerance; Immune response; Survival	Aston Sci. Inc.	2025-11-15
NCT07495124	IGFBP-2 Vaccine to Prevent Ovarian Cancer Progression in Patients With Serologic Detection of Recurrence	Not Yet Recruiting	Phase 2	26	Peptide/antigen; Whole tumor/cell lysate; DNA/RNA	Chemotherapy/SOC	Recurrent disease; Maintenance/adjuvant prevention	Immune response; Survival	University of Washington	2027-09-29
NCT07634094	Intensive Locoregional Chemotherapy, Intradermal Autologous Alpha-DC1 Vaccines, and Systemic Pembrolizumab in Treating Patients With Recurrent Ovarian Cancer	Not Yet Recruiting	Phase 2	28	Dendritic cell/DC	Adjuvant/GM-CSF; Checkpoint inhibitor; Chemotherapy/SOC	Advanced/metastatic; Maintenance/adjuvant prevention; Neoadjuvant	Safety/tolerance; Immune response; Radio-graphic response; Survival	Kalinski, Pawel, MD, PhD	2028-08-01

Table 1 (continued)

NCT Number	Study title	Current status	Phase	Enrollment	Vaccine platform	Combination strategy	Clinical setting	End-point focus	Sponsor	Primary completion date
NCT06144762	Precision Imaging for Early Detection and Targeted Treatment Monitoring in Pancreatic Cancer	Recruiting	NA	150	DNA/RNA	Not clearly specified	Basket/multiple solid tumors	Radio-graphic response	Institut du Cancer de Montpellier - Val d'Aurelle	2029-10-01
NCT05104515	First-in-human Study of OVM-200 as a Therapeutic Cancer Vaccine	Recruiting	Phase 1	36	Peptide/antigen	Not clearly specified	Basket/multiple solid tumors	Safety/tolerability; Immune response; Radio-graphic response; Bio-marker/CA-125	Oxford Vaccine Ltd. UK	2025-11-30
NCT06366490	Safety and Immunogenicity of Innocell Autologous Cellular Immunotherapy for Recurrent Epithelial Ovarian Ca...	Recruiting	Phase 1	8	Not clearly specified	Not clearly specified	Recurrent disease	Safety/tolerability; Immune response; Radio-graphic response; Bio-marker/CA-125	Photon-Pharma, Inc.	2026-11-01
NCT00722228	Autologous and Allogeneic Whole Cell Cancer Vaccine for Metastatic Tumors	Recruiting	Phase 1/ Phase 2	50	Whole tumor/cell lysate	Not clearly specified	Advanced/metastatic; Basket/multiple solid tumors	Im-mune response	Hadas-sah Medical Organization	2025-03-01
NCT05329532	Modi-1 Modiotope in Breast, Head and Neck, Ovarian, or Renal Cancer	Recruiting	Phase 1/ Phase 2	168	Peptide/antigen	Adjuvant/GM-CSF; Check-point inhibitor; Chemotherapy/SOC	Advanced/metastatic; Maintenance/adjuvant prevention; Neoadjuvant; Basket/multiple solid tumors	Safety/tolerability; Immune response; Radio-graphic response; Survival	Scancell Ltd	2026-07-01
NCT05773859	NEOadjuvant Dendritic Cell Vaccination for Ovarian Cancer	Recruiting	Phase 1/ Phase 2	10	Whole tumor/cell lysate; Dendritic cell/DC	Adjuvant/GM-CSF	Recurrent disease; Advanced/metastatic; Maintenance/adjuvant prevention; Neoadjuvant	Safety/tolerability; Immune response	Rad-boud University Medical Center	2026-03-01

Table 1 (continued)

NCT Number	Study title	Current status	Phase	Enrollment	Vaccine platform	Combination strategy	Clinical setting	End-point focus	Sponsor	Primary completion date
NCT05920798	Vaccine Therapy Plus Pembrolizumab in Treating Advanced Ovarian, Fallopian Tube, or Primary Peritoneal Cavi...	Recruiting	Phase 1/ Phase 2	40	Peptide/antigen; Whole tumor/cell lysate; Dendritic cell/DC	Checkpoint inhibitor; Chemotherapy/SOC	Recurrent disease; Advanced/metastatic	Safety/tolerability; Immune response; Radiographic response; Survival	Mayo Clinic	2028-06-30
NCT06639074	Folate Receptor Alpha Dendritic Cells (FRαDCs) or Placebo for the Treatment of Patients With Stage III or I...	Recruiting	Phase 2	78	Peptide/antigen; Whole tumor/cell lysate; Dendritic cell/DC	Not clearly specified	Recurrent disease; Advanced/metastatic; Maintenance/adjuvant prevention	Safety/tolerability; Immune response; Survival	Mayo Clinic	2027-12-31
NCT06341907	Application of MRD Combined With Personalized Vaccine in the Treatment of Postoperative Recurrence Preventi...	Recruiting	Phase 2/ Phase 3	10	Peptide/antigen; DNA/RNA	Adjuvant/GM-CSF	Recurrent disease; Maintenance/adjuvant prevention	Safety/tolerability; Radiographic response; Survival	Second Affiliated Hospital of Wenzhou Medical Univer...	2026-12-31

gradually moving from single-vaccine approaches toward combined immune modulation.

Distribution of clinical application contexts

Table 4 shows that ovarian cancer vaccine studies covered multiple clinical contexts. Advanced or metastatic disease was the most common context, involving 75 studies. Basket trials or multiple solid tumor studies involved 65 studies, recurrent disease involved 56 studies, and maintenance, adjuvant, or recurrence prevention settings involved 47 studies. In contrast, studies in platinum-resistant, platinum-sensitive, and neoadjuvant settings were relatively limited, involving 7, 4, and 5 studies, respectively.

This distribution suggests that ovarian cancer vaccine studies are currently concentrated mainly in advanced, recurrent, or multi-tumor exploratory settings, whereas evidence remains limited in perioperative, neoadjuvant, and clearly stratified platinum-sensitive or platinum-resistant populations. Endpoint emphasis also differed across clinical contexts. Advanced, recurrent, and basket trials frequently focused on safety, immune response, and radiographic efficacy. Maintenance or adjuvant settings placed greater emphasis on safety, immune durability, and survival benefit. Studies in platinum-sensitive

or platinum-resistant settings more often incorporated CA-125, radiographic response, and survival indicators.

Endpoint selection and its relationship with clinical phase

Endpoint analysis showed that safety/toxicity and immune response were the most common endpoint types in ovarian cancer vaccine studies. Figure 2B shows that safety/toxicity endpoints were mainly concentrated in Phase I and Phase I/Phase II studies, consistent with the central role of safety and tolerability assessment in early clinical trials. Immune response endpoints also occupied an important position in early-phase studies, reflecting the emphasis of cancer vaccine research on immunogenicity and mechanistic validation. In contrast, efficacy endpoints such as ORR/response, overall survival, and progression-free survival appeared more often in Phase II or later-phase studies, but their overall frequency remained limited. CA-125 and quality-of-life endpoints were relatively uncommon, suggesting insufficient use of biomarker dynamics and patient-reported outcomes in existing studies.

Table 5 further explains the clinical implications of different endpoint domains. Safety and tolerability are used mainly to evaluate adverse events, dose-limiting toxicity, and maximum tolerated dose. Immune response

Table 2 Major sponsors and research focus

Sponsor	Studies	Active/in-progress studies	Main vaccine platforms	Main clinical contexts	Example NCT IDs
National Cancer Institute (NCI)	13	1	Whole tumor/cell lysate; Peptide/antigen; Dendritic cell/DC	Basket/multiple solid tumors; Advanced/metastatic; Maintenance/adjuvant prevention	NCT00001827; NCT00004032; NCT00019916; NCT00088413
University of Washington	9	1	Peptide/antigen; Whole tumor/cell lysate; DNA/RNA	Advanced/metastatic; Basket/multiple solid tumors; Recurrent disease	NCT00005023; NCT00003002; NCT01322802; NCT07495124
Roswell Park Cancer Institute	7	0	Whole tumor/cell lysate; Peptide/antigen; Viral/vector	Recurrent disease; Advanced/metastatic; Maintenance/adjuvant prevention	NCT02432378; NCT01673217; NCT01536054; NCT01982487
Memorial Sloan Kettering Cancer Center	6	0	Peptide/antigen; Whole tumor/cell lysate; Not clearly specified	Not clearly specified; Recurrent disease; Platinum-resistant	NCT02737787; NCT02764333; NCT00006041; NCT01223235
Abramson Cancer Center at Penn Medicine	6	0	Whole tumor/cell lysate; Dendritic cell/DC	Recurrent disease; Not clearly specified; Advanced/metastatic	NCT01132014; NCT00683241; NCT01312389; NCT01312376
Gradalis, Inc.	5	1	Whole tumor/cell lysate; Not clearly specified	Basket/multiple solid tumors; Recurrent disease; Advanced/metastatic	NCT02346747; NCT01551745; NCT01867086; NCT01309230
Ludwig Institute for Cancer Research	5	0	Peptide/antigen; Viral/vector; Whole tumor/cell lysate	Basket/multiple solid tumors; Maintenance/adjuvant prevention; Recurrent disease	NCT00803569; NCT00112957; NCT00066729; NCT00623831
Mayo Clinic	5	2	Whole tumor/cell lysate; Peptide/antigen; Dendritic cell/DC	Recurrent disease; Advanced/metastatic; Basket/multiple solid tumors	NCT02111941; NCT00408590; NCT01606241; NCT05920798
Gynecologic Oncology Group	4	0	Whole tumor/cell lysate	Advanced/metastatic; Maintenance/adjuvant prevention; Recurrent disease	NCT00003386; NCT00693342; NCT00857545; NCT00551265
ImmunoVaccine Technologies, Inc. (IMV Inc.)	4	0	Peptide/antigen; Whole tumor/cell lysate; Not clearly specified	Advanced/metastatic; Maintenance/adjuvant prevention; Basket/multiple solid tumors	NCT01095848; NCT01416038; NCT03332576; NCT02785250
SOTIO a.s.	3	0	Dendritic cell/DC	Recurrent disease; Platinum-resistant; Platinum-sensitive	NCT02107378; NCT03905902; NCT03657966
University Medical Center Groningen	3	0	Peptide/antigen; DNA/RNA	Recurrent disease; Maintenance/adjuvant prevention; Not clearly specified	NCT04163094; NCT00844506; NCT04739527
Celldex Therapeutics	3	0	Peptide/antigen; Dendritic cell/DC; Not clearly specified	Advanced/metastatic; Maintenance/adjuvant prevention; Basket/multiple solid tumors	NCT00948961; NCT00648102; NCT00709462
Sumitomo Pharma America, Inc.	3	0	DNA/RNA; Peptide/antigen; Not clearly specified	Advanced/metastatic; Basket/multiple solid tumors; Maintenance/adjuvant prevention	NCT02498665; NCT01621542; NCT03311334
Michael Morse, MD	2	0	Whole tumor/cell lysate; Peptide/antigen	Maintenance/adjuvant prevention; Advanced/metastatic; Basket/multiple solid tumors	NCT00437502; NCT00027534

endpoints can indicate whether a vaccine induces antigen-specific T cells, antibody responses, or cytokine changes, but whether these responses translate into PFS, OS, or ORR benefit requires further validation. Radiographic efficacy and survival endpoints are closer to direct clinical benefit, but usually require larger sample sizes, longer follow-up, and control groups. Quality-of-life endpoints are important in maintenance therapy and recurrence prevention settings, but remain infrequently used in existing studies.

Immune monitoring reporting and assay heterogeneity

Among the 136 included trials, 92 (67.65%) reported immune monitoring or immune-response outcomes.

However, only 58 trials (42.65%) explicitly specified an assay method, whereas 34 trials (25.00%) described an immune domain without naming a specific assay; 44 trials (32.35%) did not report immune monitoring outcomes (Fig. 3A). Among the 58 trials with explicit assay reporting, 38 (65.52%) used two or more assay classes, with a mean of 2.36 explicit assay classes per explicit-assay trial (Fig. 3B).

Assay methods varied substantially across clinical phases. Frequently reported assay classes included ELISpot, cytokine/soluble-factor measurement, immune-cell phenotyping/TIL/IHC, ELISA-based assays, flow cytometry/ICS/FACS, CTL/cytotoxicity/51Cr, and DTH/skin testing, with most assay reporting concentrated in

Table 3 Conceptual interpretation of vaccine platforms

Platform	Operational definition	Representative targets/products	Typical design implication	Comment for manuscript
Peptide/anti-gen vaccine	Defined antigen or peptide target, often designed to induce antigen-specific T-cell or antibody responses.	p53, NY-ESO, folate receptor alpha, survivin, mesothelin, IGFBP-2 and other shared tumor-associated antigens.	Usually requires immune monitoring; may be combined with adjuvant, checkpoint blockade or low-dose chemotherapy.	Useful for antigen-focused trials, but patient selection and antigen expression are key translational constraints.
Whole tumor/cell ly-sate vaccine	Uses autologous or allogeneic tumor material to broaden antigen coverage.	Autologous tumor cell, tumor lysate, whole-cell vaccine and related cellular products.	Often paired with immune adjuvants, dendritic-cell loading or standard chemotherapy.	Broad antigen breadth may reduce target-miss risk, while manufacturing logistics can limit scalability.
Dendritic-cell vaccine	Ex vivo antigen presentation strategy using dendritic cells or DC-tumor fusion products.	DCVAC/OvCa, alpha-DC1, FR-alpha DC and dendritic cell/ tumor fusion approaches.	Frequently combined with GM-CSF, checkpoint blockade or locoregional chemoimmunotherapy.	Biologically rational in ovarian cancer, but batch production and immune potency remain central issues.
DNA/RNA vaccine	Nucleic-acid platform encoding tumor antigen or individualized targets.	Plasmid DNA, RNA/mRNA and personalized vaccine concepts.	Can be incorporated into minimal residual disease, adjuvant or recurrence-prevention settings.	Potentially adaptable to personalized neoantigens, but clinical maturity remains limited in this dataset.
Viral/vector vaccine	Viral delivery platform engineered to express tumor antigen or immune-stimulatory payloads.	MVA, ALVAC, vaccinia/fowlpox/ adenoviral vector approaches.	Often evaluated with immune adjuvants or checkpoint blockade.	May improve antigen delivery and immunogenicity, with vector immunity and safety monitoring as design considerations.

Phase I, Phase I/II, and Phase II studies (Fig. 3C). Biological-domain coding showed that generic immune response was the most frequent category ($n=59$), followed by T-cell/cellular response ($n=49$), humoral/antibody response ($n=28$), immune-cell phenotype/infiltration ($n=26$), and cytokine/soluble-factor response ($n=25$), whereas local/DTH response ($n=8$) and immune memory/durability ($n=4$) were less common (Fig. 3D).

Discussion

This study systematically analyzed 136 clinical trials of cancer vaccines for ovarian cancer registered in Clinical-Trials.gov as of June 2026, thereby depicting nearly three decades of clinical development in this field. Overall, ovarian cancer vaccine research can be characterized as continuous but still limited in translational progress. The number of studies did not increase linearly, but showed temporal peaks in 2011 and 2017. Clinical development was highly concentrated in early-phase studies, with Phase I trials accounting for 48.53%, whereas Phase III trials accounted for only 1.47%. Completed studies represented 55.88% of the dataset, and currently active studies accounted for 15.44%; however, only 22.79% of studies had posted results. Meanwhile, vaccine platforms were highly heterogeneous, with peptide/antigen vaccines, dendritic-cell vaccines, and whole tumor cell/lysate vaccines representing the main categories, while DNA/RNA vaccines and viral vector vaccines have gradually entered clinical exploration. Combination strategies are shifting from single-vaccine interventions toward combinations with immune checkpoint inhibitors, chemotherapy, GM-CSF, and other approaches. Clinical application contexts are concentrated mainly in advanced/metastatic

and recurrent disease, whereas evidence remains limited in perioperative, neoadjuvant, and clearly stratified platinum-sensitive or platinum-resistant populations. Endpoint selection was dominated by safety/toxicity and immune response, while clinically meaningful endpoints such as PFS, OS, and quality of life were less frequently used. These findings suggest that ovarian cancer vaccines have moved from early proof-of-concept toward translational exploration, but remain far from a clearly established clinical role.

Mechanistically, ovarian cancer has a degree of immunogenicity that provides a rationale for therapeutic vaccination. Ovarian cancer, particularly high-grade serous ovarian cancer, is frequently characterized by TP53 mutation, homologous recombination repair deficiency, and BRCA1/2-related DNA damage repair abnormalities. These molecular features influence tumor biology and may also affect antigen generation and immune recognition [1, 2]. Previous studies have shown that tumor-infiltrating lymphocytes, especially CD8 + T cells, are associated with improved prognosis in ovarian cancer, indicating that host antitumor immunity contributes to disease control [4]. However, ovarian cancer is not simply an immune-sensitive tumor. Its tumor microenvironment is also strongly immunosuppressive, with enrichment of regulatory T cells, activation of the PD-1/PD-L1 pathway, and impaired effector T-cell function [5, 17]. This state of immune recognition with constrained effector function may explain why many vaccine studies can induce antigen-specific T-cell or antibody responses but fail to consistently translate these responses into radiographic responses or survival benefits.

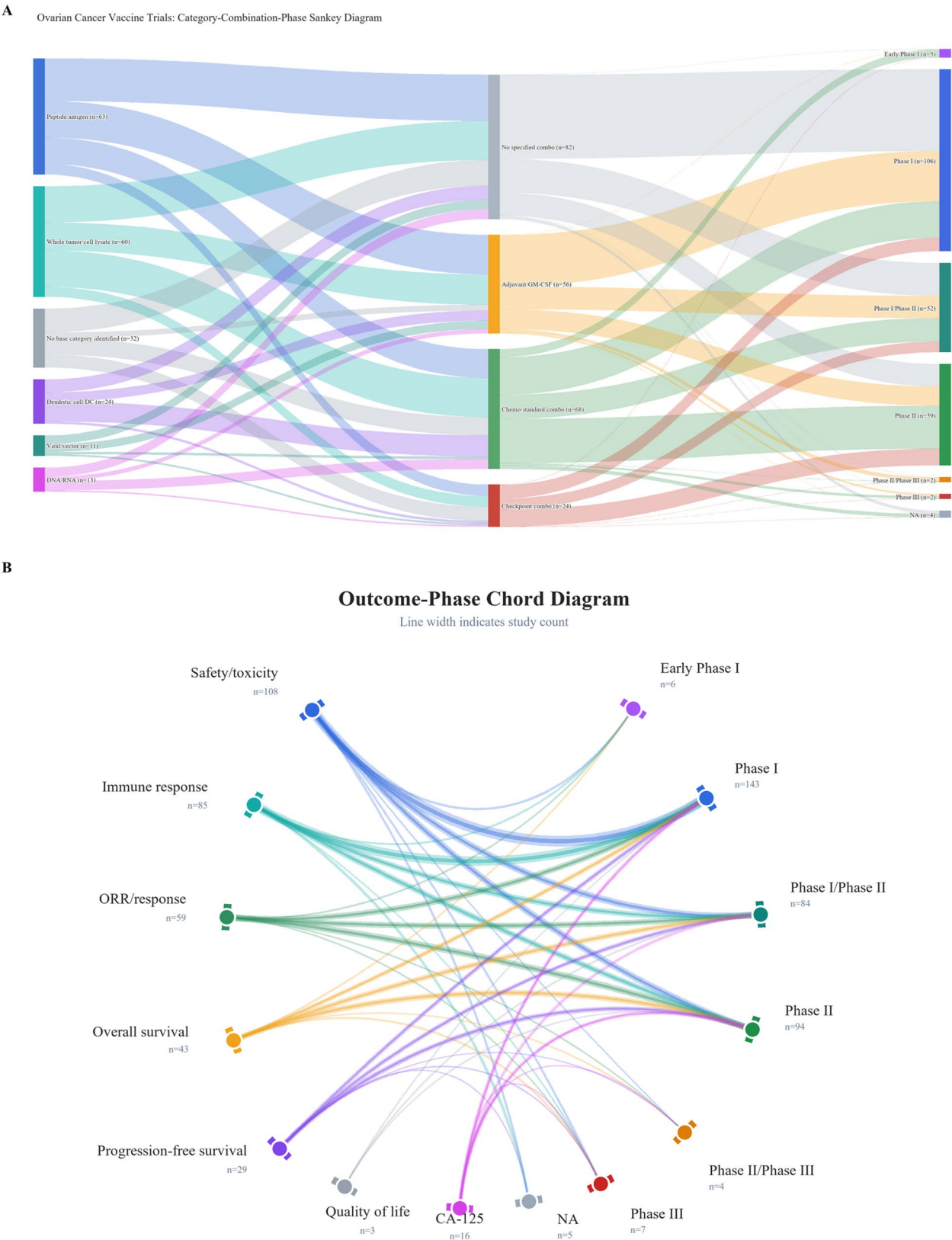


Fig. 2 (See legend on next page.)

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Fig. 2 Distribution of vaccine category, combination strategy, clinical phase, and trial outcomes. **(A)** Sankey diagram showing the relationship among vaccine category, combination strategy, and clinical trial phase. The numbers in parentheses indicate the number of corresponding study records flowing through each category; flow width is proportional to the count. **(B)** Chord diagram showing the distribution of reported outcome categories across clinical trial phases. The numbers beside each node indicate the number of outcome records in that category or phase, and line width represents the corresponding count. Because one trial could include multiple vaccine categories, combination strategies, or outcome measures, these numbers are not mutually exclusive and may exceed the total number of unique trials. ORR, objective response rate; GM-CSF, granulocyte-macrophage colony-stimulating factor; NA, not applicable

The clinical performance and translational difficulty of different vaccine platforms are closely linked to their immunological mechanisms. Peptide/antigen vaccines deliver predefined tumor-associated antigens and are relatively easy to standardize, quality-control, and monitor immunologically. However, they have limited antigen coverage and are affected by HLA restriction, heterogeneity of antigen expression, and antigen loss. Whole tumor cell or tumor lysate vaccines can provide a broader antigenic repertoire, including known and unknown antigens, and may therefore help overcome tumor heterogeneity and antigen escape, but manufacturing, batch consistency, and quality control remain major challenges. Dendritic-cell vaccines have strong immunological rationale because they enhance antigen presentation and T-cell priming. Clinical research in ovarian cancer has shown that dendritic-cell vaccination combined with bevacizumab and metronomic cyclophosphamide can induce antitumor immune responses, suggesting that combined immune modulation may enhance vaccine effects [15]. In contrast, DNA/RNA vaccines and viral vector vaccines represent evolving platforms with potential advantages in platform-based and personalized development. Personalized RNA or neoantigen vaccines have demonstrated the ability to induce neoantigen-specific T-cell responses in other tumor types, highlighting their potential as an emerging approach for individualized immunotherapy [18–20]. However, shared antigen-based vaccines have also been shown to induce antigen-specific cellular and humoral immune responses in multiple studies, and the presence of immune responses alone does not necessarily predict consistent clinical benefit [21]. Therefore, personalized neoantigen strategies should be viewed as an important complementary direction alongside shared antigen-based approaches, and their potential advantages in improving clinical outcomes require validation in prospective clinical trials.

Combination therapy may be a key route to improving the efficacy of ovarian cancer vaccines. This study found that several ongoing trials have moved from single-vaccine approaches toward combinations with immune checkpoint inhibitors, chemotherapy, or immune adjuvants. The mechanistic basis is that cancer vaccines primarily induce and expand tumor-specific T cells, whereas immune checkpoint inhibitors relieve PD-1/PD-L1- or CTLA-4-mediated suppression of T-cell function. The combination may therefore produce a priming-release

effect, in which the vaccine initiates antitumor immunity and checkpoint blockade restores effector function. However, single-agent immune checkpoint inhibitors have shown limited efficacy in ovarian cancer. KEYNOTE-100 demonstrated that pembrolizumab had a modest objective response rate in recurrent ovarian cancer, suggesting that simply relieving immune suppression is insufficient to generate consistent therapeutic effects [10]. Therefore, combinations of vaccines with checkpoint inhibitors, chemotherapy, anti-angiogenic therapy, or PARP inhibitors may be more aligned with the immunobiology of ovarian cancer. Some chemotherapeutic agents may also enhance vaccine responses by inducing immunogenic cell death, promoting antigen release, and reducing immunosuppressive cell populations, which may explain the use of low-dose cyclophosphamide in several vaccine studies.

This study has several implications for future clinical development. First, ovarian cancer vaccines may have distinct clinical applications depending on vaccine platforms and treatment contexts. Shared antigen-based vaccine approaches, including peptide/antigen vaccines targeting tumor-associated antigens such as WT1, p53, and NY-ESO-1, may be more suitable for settings with low tumor burden, including minimal residual disease, maintenance therapy, adjuvant therapy, or recurrence prevention. In these settings, vaccine-induced immune surveillance and long-term immune memory may be more likely to contribute to disease control. In recent years, PARP inhibitors and anti-angiogenic therapy have reshaped the maintenance treatment landscape for advanced ovarian cancer. SOLO1, PRIMA, and PAOLA-1 demonstrated that molecularly stratified maintenance strategies can provide durable benefits [6–8]. Therefore, cancer vaccines may represent a potential adjunctive strategy to enhance durable antitumor immunity in maintenance or recurrence prevention settings. Second, although platinum-resistant ovarian cancer often presents with recurrent disease and substantial tumor burden, it remains an important population for vaccine-based immune modulation because of limited treatment options and poor prognosis. Personalized RNA or neoantigen vaccines may provide potential advantages in this setting by enabling individualized antigen selection and addressing tumor heterogeneity. However, their clinical translation remains constrained by sequencing quality, antigen prediction accuracy, manufacturing time, cost, and accessibility [20]. Moreover, vaccines are unlikely

Table 4 Clinical contexts represented in ovarian cancer vaccine trials

Clinical context	Studies identified	Common vaccine platforms	Common endpoint focus	Representative examples
Recurrent disease	56	Whole tumor/cell lysate; Peptide/antigen; Dendritic cell/DC	Safety/tolerability; Immune response; Radiographic response	NCT02107378; NCT01334047; NCT00004032
Platinum-resistant	7	Peptide/antigen; Not clearly specified; Whole tumor/cell lysate	Immune response; Radiographic response; Survival	NCT02107378; NCT02764333; NCT01639885
Platinum-sensitive	4	Peptide/antigen; Dendritic cell/DC; DNA/RNA	Safety/tolerability; Biomarker/CA-125; Radiographic response	NCT01334047; NCT02978222; NCT04713514
Advanced/metastatic	75	Peptide/antigen; Whole tumor/cell lysate; Dendritic cell/DC	Immune response; Safety/tolerability; Radiographic response	NCT01334047; NCT02432378; NCT01456065
Maintenance/adjuvant prevention	47	Peptide/antigen; Whole tumor/cell lysate; Dendritic cell/DC	Safety/tolerability; Immune response; Survival	NCT00001827; NCT00437502; NCT06341907
Neoadjuvant	5	Whole tumor/cell lysate; Dendritic cell/DC; Peptide/antigen	Immune response; Safety/tolerability; Radiographic response	NCT02432378; NCT05773859; NCT07634094
Basket/multiple solid tumors	65	Peptide/antigen; Whole tumor/cell lysate; Not clearly specified	Immune response; Safety/tolerability; Radiographic response	NCT02346747; NCT02019524; NCT01551745

Table 5 Interpretation of common endpoint domains

Endpoint domain	Typical measures	Why it matters	Interpretive caveat
Safety/tolerability	Adverse events, dose-limiting toxicity, maximum tolerated dose, treatment discontinuation.	Essential for early-phase vaccine studies and combination regimens.	Does not establish antitumor efficacy by itself; interpretation depends on schedule and comparator.
Immune response	ELISPOT, antigen-specific T cells, antibody response, cytokines, immune-cell phenotyping.	Shows whether the vaccine engages the intended immune mechanism.	Immune activation may not translate directly into PFS, OS or objective response.
Radiographic response	RECIST-based objective response rate, disease control or response duration.	Provides a clinically interpretable anti-tumor signal in measurable disease.	May underestimate benefit for immune-based therapy if delayed response or stable disease is relevant.
Survival	Progression-free survival and overall survival.	Most clinically meaningful efficacy endpoints, especially in randomized studies.	Requires larger sample size, longer follow-up and careful control-arm selection.
Biomarker/CA-125	CA-125 kinetics, minimal residual disease markers, antigen expression or exploratory biomarkers.	Can support patient selection, early relapse detection and mechanistic interpretation.	Should usually be interpreted as supportive unless prospectively validated.
Quality of life	Patient-reported outcomes and symptom burden.	Important when vaccines are used in maintenance or recurrence-prevention settings.	Rarely central in this dataset, but useful for long-duration low-toxicity strategies.

to achieve optimal efficacy as monotherapy in advanced platinum-resistant disease and may require combination strategies, including immune checkpoint inhibitors, chemotherapy, or other immune-modulating approaches. Similarly, the development of mirvetuximab soravtansine in folate receptor alpha-positive platinum-resistant ovarian cancer highlights the importance of precise target selection in difficult-to-treat populations [9].

Endpoint selection is another central challenge in the clinical translation of ovarian cancer vaccines. This study showed that safety/toxicity and immune response were the most common endpoints, whereas ORR, PFS, OS, and quality-of-life endpoints were relatively underused. This endpoint structure is consistent with early clinical development, but limits the ability to assess true clinical value. Vaccine-induced immune responses do not necessarily equate to clinical benefit. Possible reasons include nonstandardized immune assays, differences in

sampling time points, inadequate capture of dynamic immune changes, and suppression of peripheral immune responses by the tumor microenvironment. Therefore, future trials should retain immunogenicity endpoints while more systematically incorporating composite clinical endpoints such as PFS, OS, time to recurrence, CA-125 dynamics, ctDNA, radiographic assessment, and patient-reported outcomes. Harmonized, prespecified, and vaccine-specific immune monitoring frameworks are needed to improve comparability across clinical trials and better define the relationship between immune activation, tumor control, and clinical benefit. Such frameworks should allow flexibility according to vaccine mechanisms and intended immune responses rather than relying on identical assessment strategies across all vaccine platforms.

The additional immune-monitoring analysis further supports this point. The term immune response



Fig. 3 (See legend on next page.)

(See figure on previous page.)

Fig. 3 Immune monitoring reporting and assay heterogeneity in ovarian cancer vaccine trials. **(A)** Reporting status of immune monitoring outcomes among all trials ($n = 136$). **(B)** Assay-class diversity among trials with explicit assay reporting ($n = 58$). **(C)** Heatmap of immune assay methods by clinical phase. **(D)** Biological domains of immune-response outcomes. Counts are study-level, and assay-method/domain categories are non-mutually exclusive. 51Cr assay, chromium-51 release assay; CTL, cytotoxic T lymphocyte; DTH, delayed-type hypersensitivity; FACS, fluorescence-activated cell sorting; ICS, intracellular cytokine staining; IHC, immunohistochemistry; TIL, tumor-infiltrating lymphocytes

in trial registries encompassed heterogeneous assessment approaches, including antibody/humoral assays, T-cell/cellular assays such as ELISpot and CTL assays, flow cytometry/ICS/FACS, cytokine measurements, and immune-cell phenotyping. Moreover, one quarter of all trials reported an immune domain without specifying the assay method. Future ovarian cancer vaccine trials should therefore prespecify immune-monitoring platforms, analytes, sampling time points, response thresholds, and reporting formats to improve comparability across studies and strengthen the link between immunogenicity and clinical benefit.

The main strength of this study is its comprehensive, landscape-level analysis of ovarian cancer vaccine clinical research based on ClinicalTrials.gov. The study included a relatively large number of trials over a long time span and analyzed multiple dimensions, including clinical phase, study status, geographic distribution, funding source, vaccine platform, combination strategy, clinical context, and endpoint setting. In addition, this study examined the current active pipeline and future development trends, and used visualization methods to show the complex relationships among vaccine platforms, combination strategies, and clinical phases. These analyses help clarify the overall structure of the field. However, several limitations should be acknowledged. First, ClinicalTrials.gov does not fully cover all clinical trials worldwide, and some studies may be registered in other databases, potentially leading to underestimation of non-U.S. research. Second, this was a descriptive analysis based on registry information and therefore could not directly evaluate vaccine efficacy or compare the superiority of different platforms or combination strategies. Third, registry data provide limited information on patient characteristics, immune response data, survival outcomes, and detailed treatment parameters, limiting mechanistic and efficacy-related interpretation. In addition, studies reported in the literature but not registered in ClinicalTrials.gov may have been missed, and the findings of this study will require updating as new studies are initiated and additional results become available. Overall, cancer vaccines for ovarian cancer have a clear immunological rationale and sustained translational potential. Future work should prioritize larger, randomized, multicenter Phase II/III trials with clear patient stratification and standardized endpoint design, while strengthening immune monitoring, biomarker-based selection, result disclosure, and international collaboration. Such efforts will be essential

to move the field from immunogenicity validation toward demonstrable clinical benefit.

Conclusion

Based on ClinicalTrials.gov, this study systematically characterized 136 clinical studies of cancer vaccines for ovarian cancer as of June 2026 and depicted nearly three decades of clinical development in this field. Overall, ovarian cancer vaccine research has continued to advance, but remains largely in an early exploratory stage, dominated by Phase I, Phase I/Phase II, and Phase II studies with generally small sample sizes and limited confirmatory Phase III evidence. Vaccine platforms are highly heterogeneous, including peptide/antigen vaccines, dendritic-cell vaccines, whole tumor cell or lysate vaccines, DNA/RNA vaccines, and viral/vector vaccines. At the same time, combinations with immune checkpoint inhibitors, chemotherapy, GM-CSF, and other strategies are increasing, suggesting a transition from single-vaccine interventions toward combined immune modulation. Clinical studies are concentrated mainly in advanced, metastatic, and recurrent disease, whereas evidence remains limited in perioperative, neoadjuvant, maintenance, and platinum-sensitive or platinum-resistant stratified populations. Safety and immunogenicity remain the main evaluation endpoints, while PFS, OS, objective response rate, quality of life, and biomarker-based clinical benefit endpoints are relatively underused. These findings indicate that cancer vaccines for ovarian cancer have a clear immunological foundation and translational potential, but no mature or standardized clinical application model has yet been established. Future studies should optimize patient selection, vaccine platforms, combination regimens, and endpoint design, with particular emphasis on large-scale, multicenter, randomized Phase II/III trials, immune monitoring, biomarker stratification, result disclosure, and international collaboration. Only when vaccine-induced immune responses can be reliably translated into delayed recurrence, improved survival, or better quality of life will cancer vaccines achieve a clearer role within the integrated treatment framework for ovarian cancer. The heterogeneity of immune-response assay methods and biological domains identified in this study further supports the need for pre-specified, harmonized, and vaccine-specific immune-monitoring frameworks.

Abbreviations

51Cr assay	Chromium-51 release assay
BRCA1/2	Breast cancer susceptibility gene 1/2
CA-125	Cancer antigen 125
CTL	Cytotoxic T lymphocyte
CTLA-4	Cytotoxic T-lymphocyte-associated protein 4
DC	Dendritic cell
DNA	Deoxyribonucleic acid
DTH	Delayed-type hypersensitivity
FACS	Fluorescence-activated cell sorting
FED	Federal agency
GM-CSF	Granulocyte-macrophage colony-stimulating factor
HLA	Human leukocyte antigen
ICS	Intracellular cytokine staining
IHC	Immunohistochemistry
irRC	Immune-related response criteria
iRECIST	Immune-modified response evaluation criteria in solid tumors
MHC	Major histocompatibility complex
mRNA	Messenger ribonucleic acid
NCI	National cancer institute
NIH	National institutes of health
NCT	National clinical trial identifier
ORR	Objective response rate
OS	Overall survival
PARP	Poly (ADP-ribose) polymerase
PD-1	Programmed cell death protein 1
PD-L1	Programmed death-ligand 1
PFS	Progression-free survival
RNA	Ribonucleic acid
TIL	Tumor-infiltrating lymphocytes
TP53	Tumor protein p53
WT1	Wilms tumor 1

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Use of AI

The authors declare that they have not use AI-generated work in this manuscript.

Authors' contributions

Yuping Shan: Writing – original draft, Data curation, Methodology, Visualization. Qianqian Wang: Writing – original draft, Methodology. Wei Wan: Writing – original draft, Visualization. Yan Zhang: Writing – review & editing. Yijing Chu: Writing – review & editing.

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